

Project Requirements: Mini-Trello Kanban Board

Target Audience: 7th Semester Students

Methodology: Agile / Scrum

Duration: 2 Weeks (Two 1-Week Sprints)

1. Project Overview

The goal of this project is to build a simple, single-page Kanban board where users can create tasks and move them between three columns: **To Do**, **In Progress**, and **Done**. This project serves as a practical application of Frontend development, Backend API creation, Database management, and Agile methodologies.

2. Technical Requirements

- **Frontend:** HTML, CSS, JavaScript (React, Vue, or Vanilla JS acceptable).
- **Backend:** Node.js (Express), Python (Django/Flask), or Java (Spring Boot).
- **Database:** MongoDB, MySQL, or PostgreSQL.
- **Architecture:** RESTful API design.

3. UI / Layout Guidelines

The user interface must be clean, responsive, and follow this structure:

- **Header:** Display the project title ("Mini-Trello") and a prominent "+ Create New Task" button.
- **Main Board Layout:** A 3-column grid representing the status of tasks.
 - Column 1: **To Do** (Suggested background: Light Gray)
 - Column 2: **In Progress** (Suggested background: Light Blue)
 - Column 3: **Done** (Suggested background: Light Green)
- **Task Cards:** Each task should be displayed as a distinct card containing the task title, description, and action buttons (Next/Previous) to move the card between columns. Alternatively, Drag-and-Drop functionality can be implemented.

4. Initial Product Backlog

The team must manage these User Stories in their Agile backlog, estimate effort (story points), and pull them into active sprints.

Priority	User Story	Acceptance Criteria
High	As a user, I want to create a new task so that I can add it to my board.	Clicking "Add Task" opens a form. Form requires Title and Description. Submitting saves to DB and adds to "To Do".
High	As a user, I want to see all my tasks grouped by status so I know what to work on.	Page loads 3 columns. API fetches tasks from DB. Tasks render in correct columns based on their status.
High	As a user, I want to move a task from "To Do" to "In Progress" to "Done".	Card has buttons (or drag/drop) to change status. Triggers API update. UI reflects change immediately.
Medium	As a user, I want to delete a task if it was created by mistake.	Card has a delete icon. Prompts confirmation. Deletes from DB and UI.

5. Agile Execution & Sprint Breakdown

The team will operate in two one-week sprints.

Sprint 1 Goal: Foundation & Setup

1. **Agile Ceremonies:** Conduct Sprint Planning. Assign tasks to team members.
2. **Database Setup:** Design the schema for tasks (Fields: id, title, description, status).
3. **Backend API:** Create routes for Create and Read (GET all tasks, POST new task).
4. **Frontend Layout:** Build the static HTML/CSS structure with 3 columns.

Deliverable at Sprint 1 Review: A working database, API endpoints that can be tested via Postman, and a static frontend UI.

Sprint 2 Goal: Integration & Delivery

1. **Agile Ceremonies:** Conduct Sprint 1 Retrospective, Sprint 2 Planning.
2. **Backend API:** Complete the Update and Delete routes (PUT/PATCH status, DELETE task).
3. **Frontend Integration:** Connect the Frontend UI to the Backend API using fetch or Axios.
4. **Interactivity:** Ensure clicking/moving a card triggers the correct API call and updates the UI dynamically.

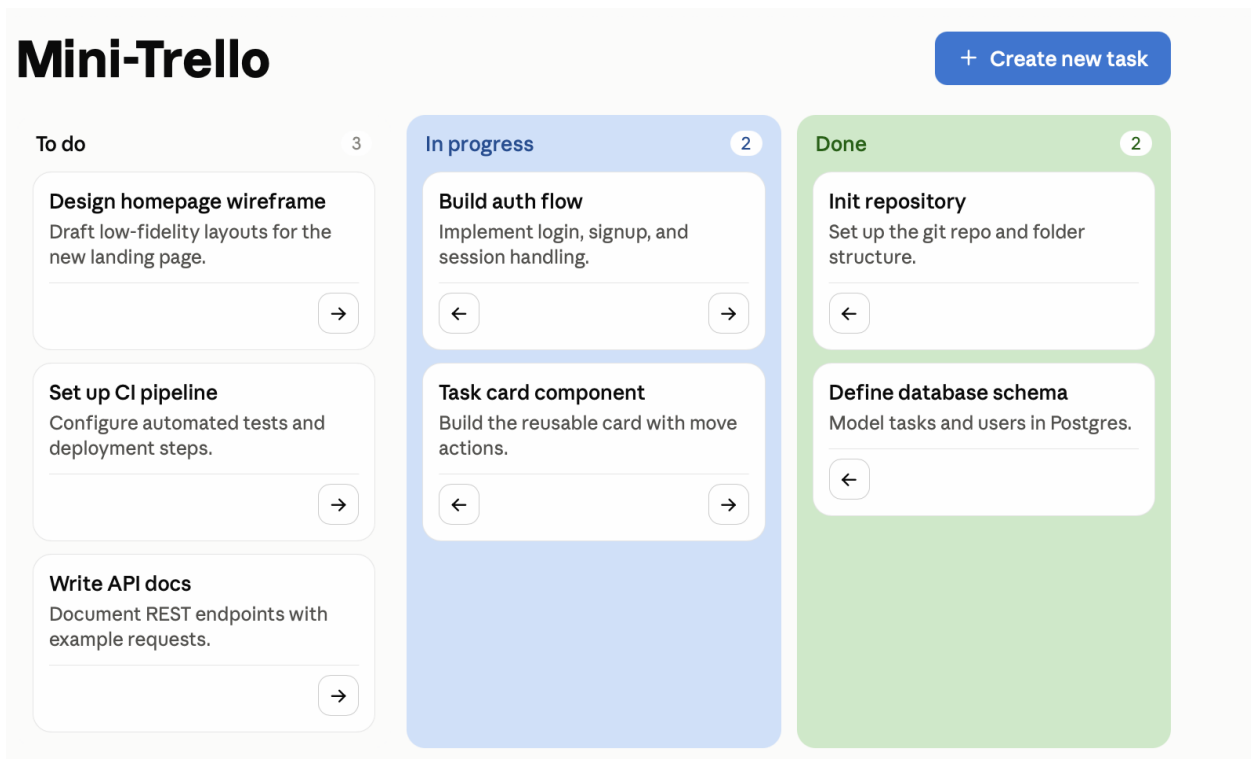
Deliverable at Sprint 2 Review: A fully functional Mini-Trello board where actions on the UI correctly persist in the database.

6. API Payload Example (Reference)

Below is the standard JSON structure expected for a Task object:

```
{
  "id": "101",
  "title": "Design Database Schema",
  "description": "Create ER diagram for the task and user tables.",
  "status": "in_progress",
  "assigned_to": "Sarah"
}
```

7. Design Example (Reference)



The above given image is for reference try your own design but all sprints should be satisfied

8. Deliverable & Submission instructions

Students must submit two mandatory files prior to the deadline: a comprehensive project report document and the compressed project source code.

Required Submission Items:

- **Project Report Document (.docx format):**
 - **Student Details:** Name, Enrollment Number and Semester.
 - **Screenshots:** Clear, high-resolution screenshots showing your running "Mini-Trello" application UI (Header, Task Modal, and 3-column board layout with active cards).

- **Agile Execution Summary:** Proof of Sprint 1 & Sprint 2 completions, User Stories addressed, and team responsibility breakdown.
- **API Endpoints Table:** A summary list of developed REST endpoints (GET, POST, PUT/PATCH, DELETE) and sample request/response payloads.
- **Source Code Archive (.zip or .rar format):**
 - A compressed file containing both Frontend and Backend codebases.
 - Include a **README.md** file with step-by-step instructions to run the application locally.
 - Exclude heavy dependency folders (such as **node_modules** or **venv**)—ensure dependency requirements are properly listed in **package.json** or **requirements.txt**.